Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b

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Editorial Decision

MINOR REVISION

Weighted Score: 3.7 / 5.0
Confidence: 82%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**5.0**	Excellent	■■■■■
Manuscript Structure	**2.5**	Weak	■■■■■
Methodology & Equations	**3.5**	Good	■■■■■
Statistical Analysis	**4.5**	Excellent	■■■■■
Results & Evidence	**4.5**	Excellent	■■■■■
Discussion & Conclusions	**3.0**	Acceptable	■■■■■
Figures & Tables	**3.5**	Good	■■■■■
Scientific Writing	**3.5**	Good	■■■■■
References & Citations	**3.0**	Acceptable	■■■■■
Ethics & Transparency	**2.0**	Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **5.0 / 5.0** | Confidence: 95%

Strengths:

- [TITLE] is specific and informative, clearly reflecting the contribution of the physics-informed EANN framework.
- [ABSTRACT] follows a structured format with clear background, objective, methods, results, and conclusions.
- [KEYWORDS] include a mix of specific and general terms, with domain-specific indexing terms.
- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.

StructureReviewer

Manuscript structure and logical coherence

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear title and abstract
- Well-defined methodology section
- Experimental results presented

Weaknesses:

- Missing literature review section
- Redundant content in sections_vi and section_vii
- Research question/hypothesis not explicitly stated

Major Comments:

- The manuscript lacks a literature review section, which is critical for establishing context and research gap.
- Sections_vi and section_vii are duplicated and should be merged or removed.
- The research question or hypothesis is not explicitly declared in the introduction.

Minor Comments:

- Section map includes non-standard sections like 'preamble' and 'biography' which may confuse readers.
- Some sections have incomplete content (e.g., introduction, experiments).

Recommendations:

- Add a literature review section to contextualize the research.
- Merge or remove duplicated sections_vi and section_vii.
- Explicitly declare the research question or hypothesis in the introduction.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear integration of physical constraints into the optimization framework
- Comprehensive statistical validation using nonparametric tests
- Well-defined experimental setup with a physical prototype

Weaknesses:

- Lack of detailed description of the EANN architecture and training process
- Limited discussion on the choice of multi-objective optimization algorithms

- Insufficient detail on the implementation of the evolutionary search strategy

Major Comments:

- The paper lacks a detailed description of the EANN architecture and training process, which is critical for reproducibility and understanding of the method.

Minor Comments:

- Equations are not consistently introduced or referenced in the text, which may hinder clarity.

Recommendations:

- Provide a detailed description of the EANN architecture, including network structure, activation functions, and training procedure.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **4.5 / 5.0** | Confidence: 80%

Strengths:

- Use of Kruskal-Wallis and post-hoc Wilcoxon tests for non-normal data
- Bootstrap confidence intervals for robustness
- Coefficient of variation and RMSE/MSE for practical significance
- Pareto front and hypervolume indicators for multi-objective analysis

Weaknesses:

- Lack of p-values for statistical tests
- No explicit justification for 30 runs as sufficient
- No mention of bootstrap resample size or variance
- No explicit details on how Holm correction was applied

Major Comments:

- The manuscript should report p-values for Kruskal-Wallis and post-hoc Wilcoxon tests to fully support statistical significance claims.

Minor Comments:

- Clarify the justification for using 30 independent runs.
- Specify the number of bootstrap resamples used for confidence intervals.

Recommendations:

- Report p-values for all statistical tests to ensure transparency.
- Provide details on the number of bootstrap resamples and variance for confidence intervals.
- Clarify how the Holm correction was applied to post-hoc comparisons.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Equations are numbered sequentially and referenced in the text
- Figures and tables are referenced in the text before they appear
- The methodology is clearly described and aligns with the figures and tables

Weaknesses:

- Some figures lack axis labels and units, reducing their self-explanatory quality
- Some tables lack units, statistical significance indicators, and explicit metric labels
- Some equations are referenced but not fully defined in the text

Major Comments:

• Equation (11) is incomplete and should be fully defined in the text
• Figure 3 and 4 lack axis labels and units, which reduces their clarity and self-explanatory nature
• Table X lacks units, statistical significance indicators, and explicit metric labels

Minor Comments:

• Figure 5 does not specify units for stiffness and damping parameters
• Figure 6 does not clearly label the axes or provide a legend for the Pareto front
• Equation (13) is referenced but not fully defined in the text

Recommendations:

• Add units to the axes of Figures 3, 4, and 5
• Fully define Equations (11), (13), and (16) in the text
• Clarify the metric labels in Table X and add units and statistical significance indicators

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ResultsReviewer

Results validity and empirical support

Score: **4.5 / 5.0** | Confidence: 80%

Strengths:

• Quantitative metrics (MSE, RMSE, CV, hypervolume) are reported and tied to figures/tables
• Baseline comparisons with GA and PSO are explicitly included
• Statistical validation methods (Kruskal-Wallis, Wilcoxon, bootstrap) are described

Weaknesses:

• Negative results or limitations of the EANN framework are not explicitly acknowledged
• The Pareto front analysis is mentioned but not fully detailed or tied to specific figures/tables
• The 30 independent runs are mentioned but not clearly connected to the variability of results

Major Comments:

• The manuscript lacks explicit acknowledgment of limitations or negative results, which is critical for a complete evaluation.

Minor Comments:

• The Pareto front analysis is referenced but not sufficiently detailed in the results section.

Recommendations:

• Include a dedicated subsection for limitations and negative results to meet the Q1 rubric requirements.
• Clarify how the 30 independent runs are used to assess variability and consistency of results.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.0 / 5.0** | Confidence: 80%

Strengths:

• Clear experimental validation and comparison with baselines
• Contribution to real-time structural health monitoring
• Statistical significance of results

Weaknesses:

• Discussion lacks comparison with prior work
• Overgeneralization of results
• Future work is not specified

Major Comments:

• The Discussion section lacks a meaningful comparison with prior work in the field, which is critical for establishing the novelty and relevance of the findings.

Minor Comments:

- The conclusion section could benefit from more specific quantification of results.

Recommendations:

- Include a detailed comparison with prior work in the Discussion section to contextualize the findings.
- Avoid overgeneralization by specifying the scope and limitations of the results.
- Provide concrete future work directions that are justified by the results.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical descriptions of the EANN framework
- Robust statistical validation methods
- Well-structured experimental setup

Weaknesses:

- Overuse of passive voice
- Lack of definition for acronyms
- Vague quantifiers without numerical support

Major Comments:

- The manuscript contains several instances of passive voice that could be rephrased for clarity and directness.
- Acronyms like EANN, PSO, and GA are used without initial definition, which may confuse readers.
- The use of vague quantifiers such as 'significant' and 'many' should be replaced with specific numerical values for precision.

Minor Comments:

- Some sentences are overly complex and could be simplified for better readability.
- A few sentences could benefit from more precise connectors for smoother transitions.

Recommendations:

- Define all acronyms on first use.
- Replace vague quantifiers with numerical values.
- Revise passive voice constructions to active voice where appropriate.
- Simplify complex sentences for improved clarity.

ReferencesReviewer

References quality, recency, and relevance

Score: **3.0 / 5.0** | Confidence: 80%

Strengths:

- Citations are consistent in IEEE format
- Recent references from 2021-2023 are included

Weaknesses:

- Lack of seminal foundational works in structural dynamics
- Missing key areas in literature review

Major Comments:

- The references section lacks seminal foundational works in structural dynamics and seismic design regulations, which are central to the manuscript's context

Minor Comments:

- Some references, like [29], are not directly tied to the methodology or claims made in the introduction

Recommendations:

- Include seminal works on structural dynamics and seismic design regulations
- Ensure all key areas mentioned in the introduction are supported by relevant references

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EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **2.0 / 5.0** | Confidence: 80%

Strengths:

- Limitations are explicitly stated in the discussion section

Weaknesses:

- Limitations are not specific or detailed, which reduces their credibility
- Reproducibility is not addressed, which is critical for scientific validation

Major Comments:

- The manuscript lacks essential ethical reporting, including ethical approval, data availability, conflict of interest, and funding disclosure. These are required for transparency and reproducibility.

Minor Comments:

- Limitations are mentioned but not sufficiently detailed to assess their impact on the conclusions.

Recommendations:

- Authors should provide a detailed data availability statement and disclose any potential conflicts of interest.
- Include a specific discussion of limitations, their impact on the study's conclusions, and how they affect the generalizability of the findings.
- Clarify the reproducibility of the study by providing access to data and code.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] The manuscript lacks a literature review section, which is critical for establishing context and research gap.

[2] Sections_vi and section_vii are duplicated and should be merged or removed.

[3] The research question or hypothesis is not explicitly declared in the introduction.

[4] The paper lacks a detailed description of the EANN architecture and training process, which is critical for reproducibility and understanding of the method.

[5] The manuscript should report p-values for Kruskal-Wallis and post-hoc Wilcoxon tests to fully support statistical significance claims.

[6] Equation (11) is incomplete and should be fully defined in the text

[7] Figure 3 and 4 lack axis labels and units, which reduces their clarity and self-explanatory nature

[8] Table X lacks units, statistical significance indicators, and explicit metric labels

[9] The manuscript lacks explicit acknowledgment of limitations or negative results, which is critical for a complete evaluation.

[10] The Discussion section lacks a meaningful comparison with prior work in the field, which is critical for establishing the novelty and relevance of the findings.

[11] The manuscript contains several instances of passive voice that could be rephrased for clarity and directness.

[12] Acronyms like EANN, PSO, and GA are used without initial definition, which may confuse readers.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] Section map includes non-standard sections like 'preamble' and 'biography' which may confuse readers.

[2] Some sections have incomplete content (e.g., introduction, experiments).

[3] Equations are not consistently introduced or referenced in the text, which may hinder clarity.

[4] Clarify the justification for using 30 independent runs.

[5] Specify the number of bootstrap resamples used for confidence intervals.

[6] Figure 5 does not specify units for stiffness and damping parameters

[7] Figure 6 does not clearly label the axes or provide a legend for the Pareto front

[8] Equation (13) is referenced but not fully defined in the text

[9] The Pareto front analysis is referenced but not sufficiently detailed in the results section.

[10] The conclusion section could benefit from more specific quantification of results.

[11] Some sentences are overly complex and could be simplified for better readability.

[12] A few sentences could benefit from more precise connectors for smoother transitions.

Specific Improvement Recommendations

1. Add a literature review section to contextualize the research.
2. Merge or remove duplicated sections_vi and section_vii.
3. Explicitly declare the research question or hypothesis in the introduction.
4. Provide a detailed description of the EANN architecture, including network structure, activation functions, and training procedure.
5. Report p-values for all statistical tests to ensure transparency.
6. Provide details on the number of bootstrap resamples and variance for confidence intervals.
7. Clarify how the Holm correction was applied to post-hoc comparisons.

8. Add units to the axes of Figures 3, 4, and 5
9. Fully define Equations (11), (13), and (16) in the text
10. Clarify the metric labels in Table X and add units and statistical significance indicators
11. Include a dedicated subsection for limitations and negative results to meet the Q1 rubric requirements.
12. Clarify how the 30 independent runs are used to assess variability and consistency of results.

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